Assembly Guide

This guide gives an overview on the steps to assemble the K12 STEM Experience mats.

From the shipment, the following is included:

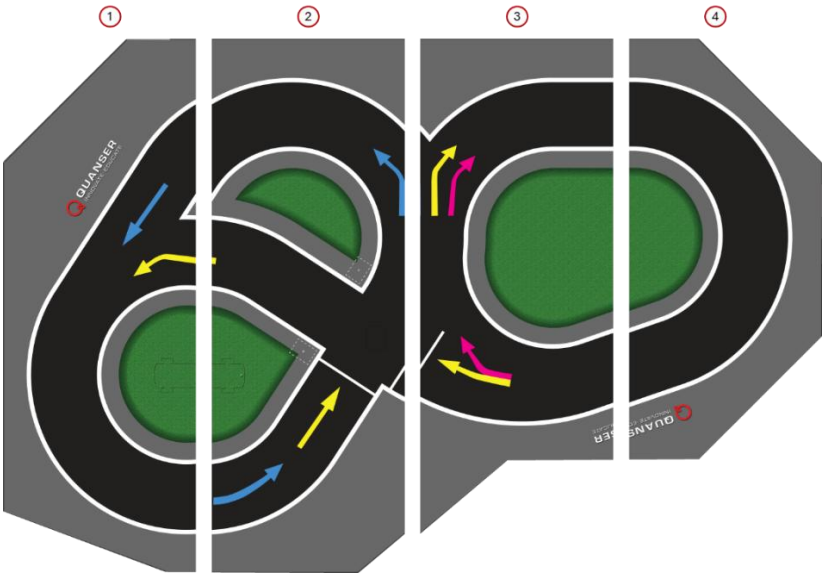
- 2x roll of STEM mat segments
- 13x white walls
- 2x traffic lights
- 2x traffic lights power cable
- 2x traffic lights power bank
- 3x QCar
- 3x QCar RGBD camera cables
- 6x QCar battery
- 3x QCar battery charger
- 26x white wall adjoining caps
- 26x white wall feet
- 1x bag of road signs
- 1x TP link router
- 1x YOGA laptop

The packaging should include a few extra feet and caps for assembly. Don't worry if there are extra pieces because of this.

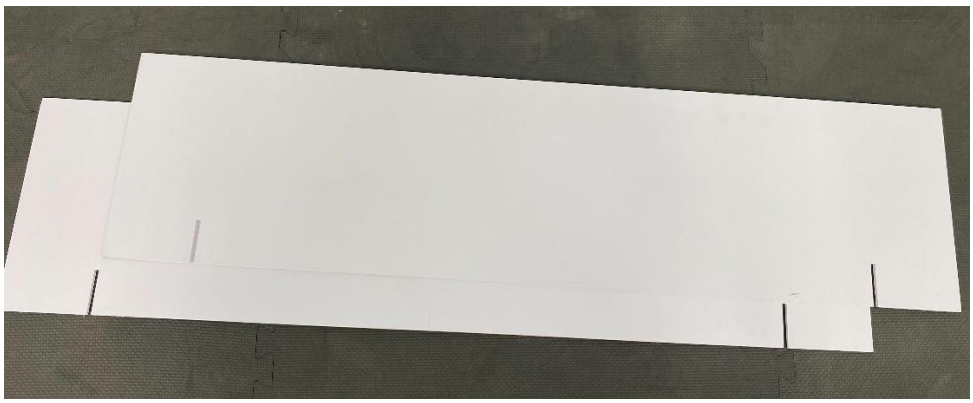
Additionally, the following peripherals will need to be placed on the mat:

- 2x traffic lights
- 2x power bank for traffic lights
- 3x QCar

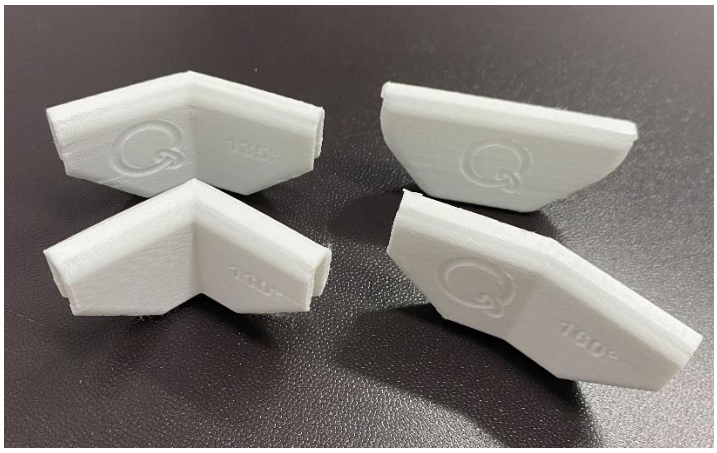
Diagram of mats and major components



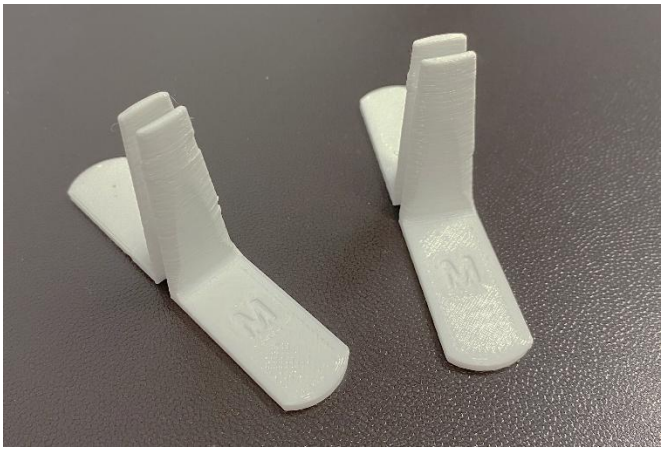
STEM Mat Segment 1-4



White Walls



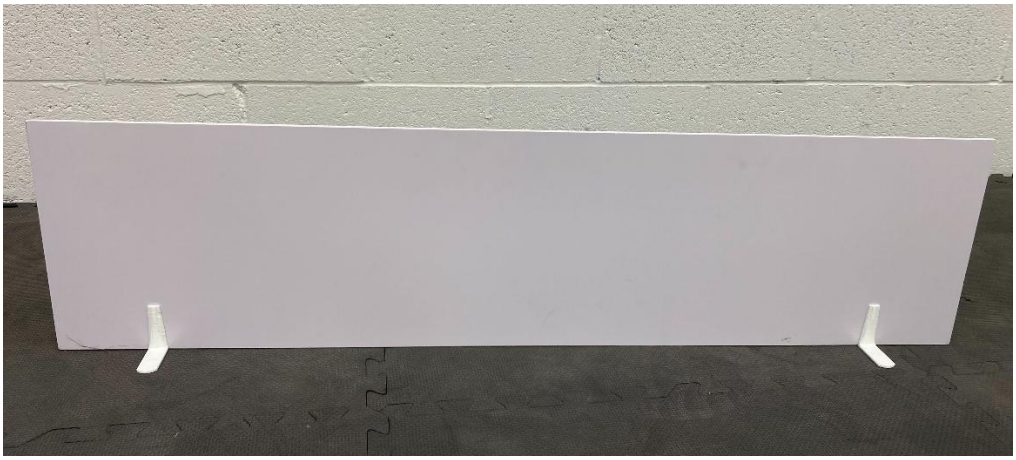
Wall Adjoining Caps, Marked by Angles (110˚, 135˚, 160˚, 180˚)



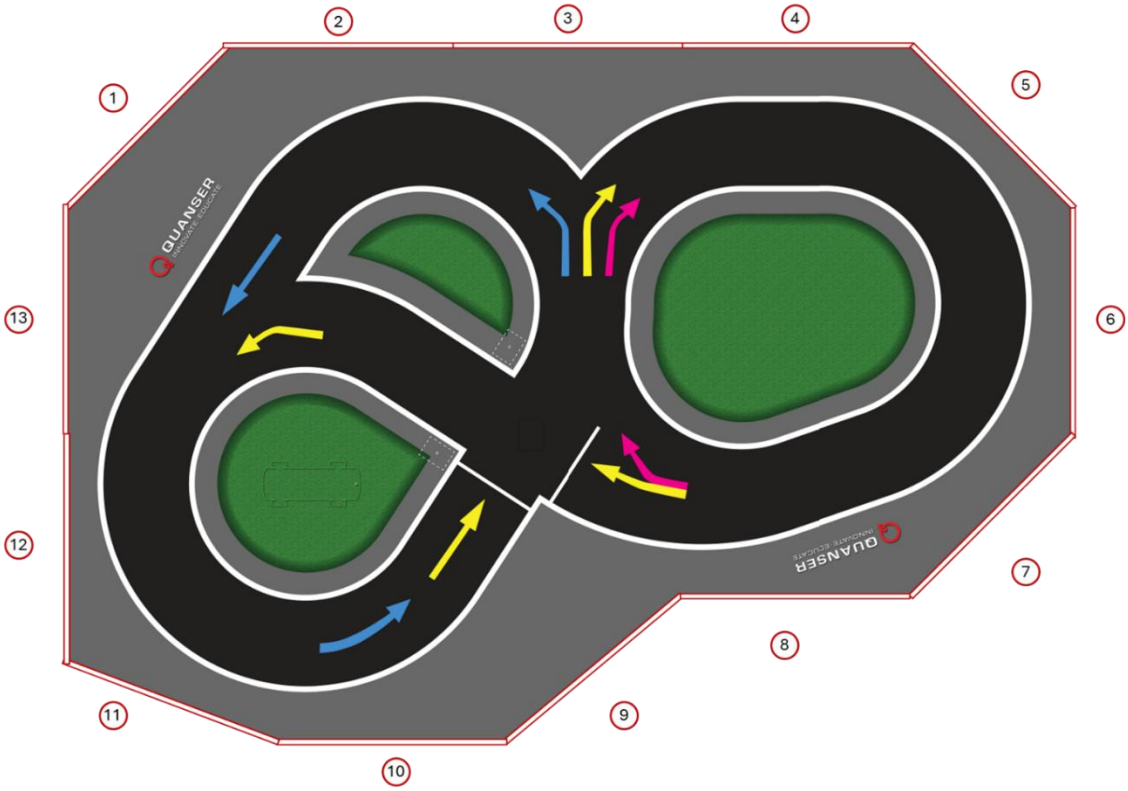
Wall Feet, Marked with "M" on One Side

Assemble the STEM Mats

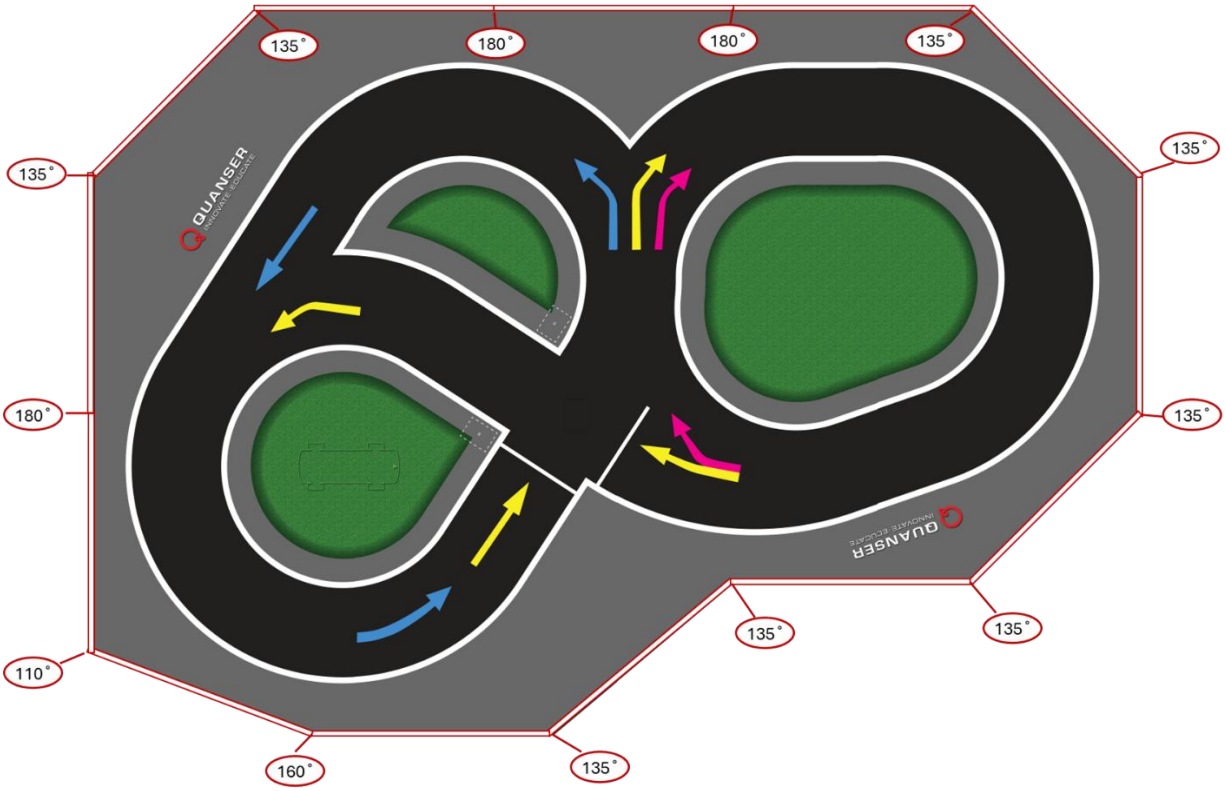
1. Attach 2 wall feet to the slots in the white wall. Make sure “M” side of the two feet are on the same side of the wall. An assemble wall is shown below.

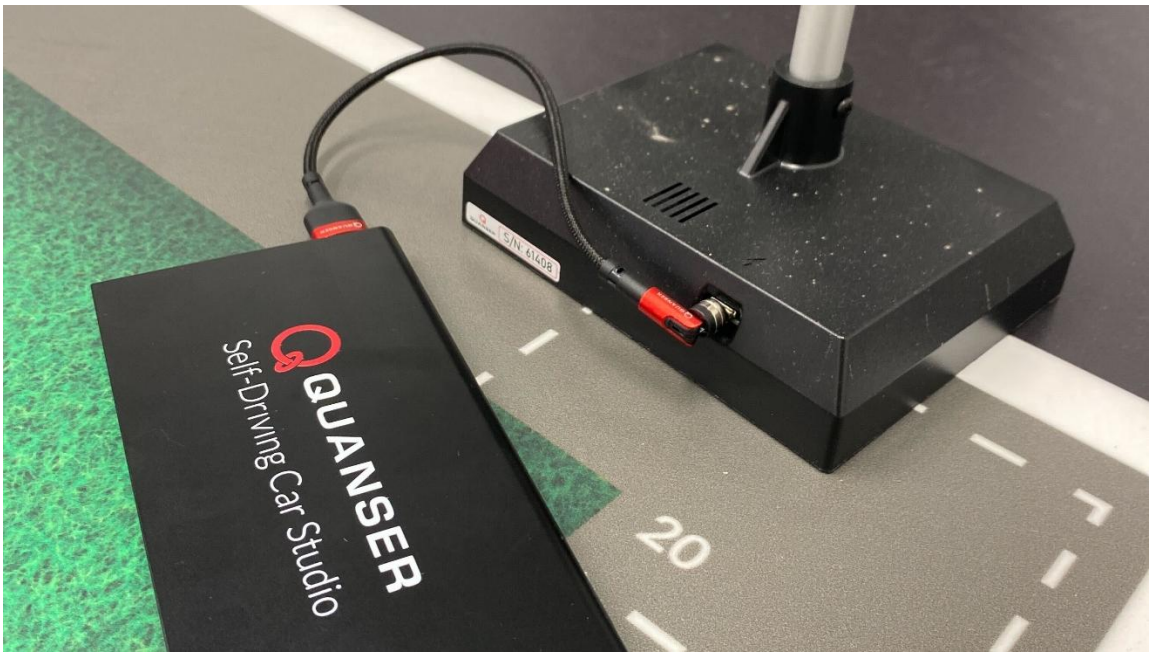


2. Lay down the mats together and place lace the wall segments against the mats. Make sure the side marked with “M” faces the mat, and the leg is tucked under the mat. The 13 walls are of equal length, and their placement is shown below.



3. Four variations of the wall adjoining caps are provided, as shown in the earlier section. The angles between neighbouring walls are shown below.



4. Use the proper adjoining caps to connect the walls. Two adjoining caps are needed for each connection (top and bottom). An example of two walls connected with 135° is shown below.

5. Connect the traffic lights to the power bank using the magnetic power and place them at marked outline on the mat (20 or 21 based on the IP address of the traffic light). User may find instructions on reserving IP addresses to traffic lights in Operating Manual (*E. Setting Up -> Router Setup*).